MIL-E-1/15208(EC)
12 August 1976
SUPERSEDING
MIL-E-1/1520A(NAVY)
26 February 1965
(See note 25)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 8196

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Amplifier pulsed, broadband, 3,400 to 3,600 MHz, integral cavities, electromagnetic focusing, integral Vacion pump

ABSOLUTE RATINGS:

Parameter:	Ef	if(surge)	tk	(focus I each coil)	epy	epx	ik	Ik
Unit:	V	a	Minutes	A	kv	kv	a	mAdc.
Maximum:	8.5	54	---	8	125	40	100	350
Minimum:	---	---	15	4	---	---	---	---
		(Note 1)		(Note 2)	(Note 3)			

ABSOLUTE RATINGS:

Parameter:	pi	Pi	tp	Du	pd	Pd	Load	Waveguide pressurization (dry air)
Unit:	Mw	kW	us	---	kw	W	VSWR	psia
Maximum:	12.5	40	13	0.004	1.5	20	1.5:1	50
Minimum:	---	---	---	---	---	---	---	40

ABSOLUTE RATINGS:

Parameter:	Temperature		Coolant flow water	Hydrostatic pressure	Ambient temperature	
	Cathode seal	Coolant inlet			Operating	Storage
Unit:	°C	°C	GPM	psig	°C	°C
Maximum:	150	50	---	150	69	85
Minimum:	---	10	10	---	-29	-62
	(Note 4)	(Note 5)	(Note 5)			(Note 6)

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	RF input:	Series N coaxial panel jack (see figure 1)
Base:	Special (see figure 1)		
Mounting position:	Vertical, cathode down (see note 7)		
Cooling:	Distilled water (see notes 5 and 8)	RF output:	Waveguide flange (see figure 1)
Cathode:	Unipotential, oxide coated		
Focusing:	Electromagnet (see note 9)		

TEST CONDITIONS:

Parameter:	Ef	tk	I (focus each coil)	epy	tp(rf)	Du(rf)	pd	Load VSWR	Coolant flow water
Unit:	V	Minutes	A	kv	us	---	w	---	GPM
Tolerance:	---	Min	---	---	---	---	Max	Max	Min
	(Note 24)	15	(Note 2)	115	12	0.0036	500	1.2:1	10

GENERAL:

First article sample inspection - Required (see note 10)
Service-life guarantee - See note 11

Frequency	
F	MHz
F1	3,400
F2	3,430
F3	3,570
F4	3,600

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	First article sample inspection						
1042	Shock, specified pulse	12	No voltages applied; amplitude = 10 G	---	---	---	---
1031	Low-frequency vibration	13		---	---	---	---
1011	Humidity	14		---	---	---	---
---	Input VSWR	-	Power output (1), and (2)	d	---	6:1	VSWR
	Quality conformance inspection, part 1						
---	Pressure drop	-	Waterflow = 10 GPM	ΔP	---	55	psi
4289	Heater current (nonoperating)	24		If	20	38	A
4250	Power output (1)	5, 15, 16, 18	F2 to F3 in 10 MHz steps; epy = 125 kv (max)	po	2.7	3.7	Mw
4250	Power output (2)	5, 15, 16, 18	F1 to F2 and F3 to F4 in 10 MHz steps; epy = 125 kv (max)	po	2.0	3.7	Mw
1256	Electrode current (cathode)	-	pd = 0	Ik	70	86	A
4214	Cathode emission	17		$\frac{\Delta Ik}{Ik}$	---	10	%
4006	X-radiation	19		---	---	3	mR/hr
	Quality conformance inspection, part 2						
4266	Direct-interelectrode capacitance	20		Cka	---	30	pF
4003	Pressurizing	21	Method A; 50 psia; t = 15 minutes	---	---	---	---
1105	Permanence of marking	22		---	---	---	---
	Quality conformance inspection, part 3						
---	Service life	11	Group D	t	1,000	---	hrs
---	Life-test end point:						
---	Power output (1)	-		po	2.1	---	Mw

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NOTES:

1. The heater voltage shall be applied such that the maximum surge current does not exceed the specified ratings.

2. The magnetic field current, for each of the four coils, shall be adjusted with ±10 percent of the value stamped on the tube's name plate. If only three magnet power supplies are used, coils No. 2 and No. 3 shall be connected together in series and coils No. 1 and No. 4 connected separately. The current in the series shall be within ±10 percent of the average value of No. 2 and No. 3 coil current readings on the name plate. Some degradation of performance may be experienced when operating focusing magnet with only three power supplies. Beam voltage shall not be applied until focusing currents are adjusted, and precautions shall be taken to prevent improper magnetic fields from adversely affecting the tube's performance.

3. Short transient spikes on the voltage pulse shall not exceed 140 kv.

4. Temperature measurements shall be made at the location shown on figure 1. The cathode seal shall be completely immersed in oil within the limits shown on figure 1.

5. The waterflow rate shall be 10 GPM at inlet temperatures between +10° and +50°C. The water temperature at the inlet of the collector shall never exceed +50°C.

6. Before placing tube in storage, the coolant shall be removed to avoid damage to the tube by blowing warm air through the coolant passages.

7. The TUT shall be mounted with its beam axis vertical. A flexible length of waveguide or some other suitable connection shall be provided between the TUT output flange and load or output waveguide system to prevent excessive forces from being applied to the output flange.

8. The TUT shall be cooled with distilled water. The purity of the distilled water shall be maintained during the operation of the system. Corrosive and scaling conditions shall be avoided.

9. Electromagnet, Varian type VA-1512B, or equivalent.

10. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this tube specification sheet. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previous procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

11. Service life of the tube shall be 1 year or 1,000 operating hours (beam voltage). Life test shall be performed only when specified in the contract or order.

12. The TUT shall be tested in a standard drop type shock testing machine such as a Component Shock Machine or equivalent. The TUT shall be shocked three times in each of three mutually perpendicular planes, one of which shall be parallel to the beam axis. After this test, the TUT shall meet the requirements of the power output (1) and (2) tests.

13. No voltages shall be applied during vibration test. The tube shall be mounted rigidly on a table vibrating with simple harmonic motion with the tube in position X. The tube shall be vibrated at the specified frequency and amplitude for 1 minute. Following vibration test, the TUT shall pass requirements of power output (1) and (2) tests.

14. TUT shall be subjected, nonoperating, to an atmosphere of 90 to 100 percent relative humidity at a temperature of +95° to +100°C for a period of 96 hours. The TUT shall not show mechanical failures, harmful corrosion, loss of plating, paint, and so forth, or any other defects or deterioration which may interfere with its operation. After this test, the TUT shall be tested for power output (2) at F4 only.

15. Focusing coil currents and incident drive power shall be adjusted as specified on individual tube name plate (and test performance sheet). No further adjustments will be required within the frequency rahge specified.

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NOTES: -Continued

16. The specified output power levels shall be obtained with a constant rf drive level not to exceed the specified maximum.

17. When operated under the specified cathode emission test conditions, the heater voltage (Ef) shall be decreased by 10 percent from the name plate value and the cathode emission allowed to stabilize. The percent change in peak cathode current shall not exceed the value specified (10 percent).

$$\frac{\Delta Ik}{Ik}\% = \frac{Ik_1 - Ik_2}{Ik_1} \times 100$$

Ik_1 - Peak stabilized cathode current in amps obtained with Ef = name plate value.

Ik_2 = Peak stabilized cathode current in amps obtained with Ef = 90% name plate value.

18. The TUT shall satisfy the peak power requirements specified herein. The optimum beam voltage (epy) chosen for optimum power output performance of the individual tube shall be marked on the tube data sheet.

19. Upon request, typical data will be provided of X-ray measurements. No measurements will be taken in the area between the plane of magnet to oil tank interface and bottom of cathode. Partial X-ray shielding is provided. It is the responsibility of the equipment designer to insure that radiation levels are compatible with existing federal and military requirements. At normal voltages there will be less than 3 mr per hour at 3 feet from centerline of tube and magnet. The only significant radiation is in the area of the holes where the cable, cooling hose, and waveguide come through the shielding.

20. The anode to cathode capacitance shall be measured in air without heater and beam voltage leads connected.

21. The output transmission line shall show no indication of leaks.

22. A name plate shall be permanently affixed to the tube in a conspicuous location, either on the top of the upper pole piece or at any convenient location above this surface and shall include the following information: tube manufacturer, tube type number, tube serial number, incident drive power, proper operating current for each solenoid section, and proper filament voltage.

23. The O-ring specified on figure 1 shall be supplied with the tube.

24. Ef = Name plate value.

25. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

User activities:
Navy - AS, OS, MC, CG

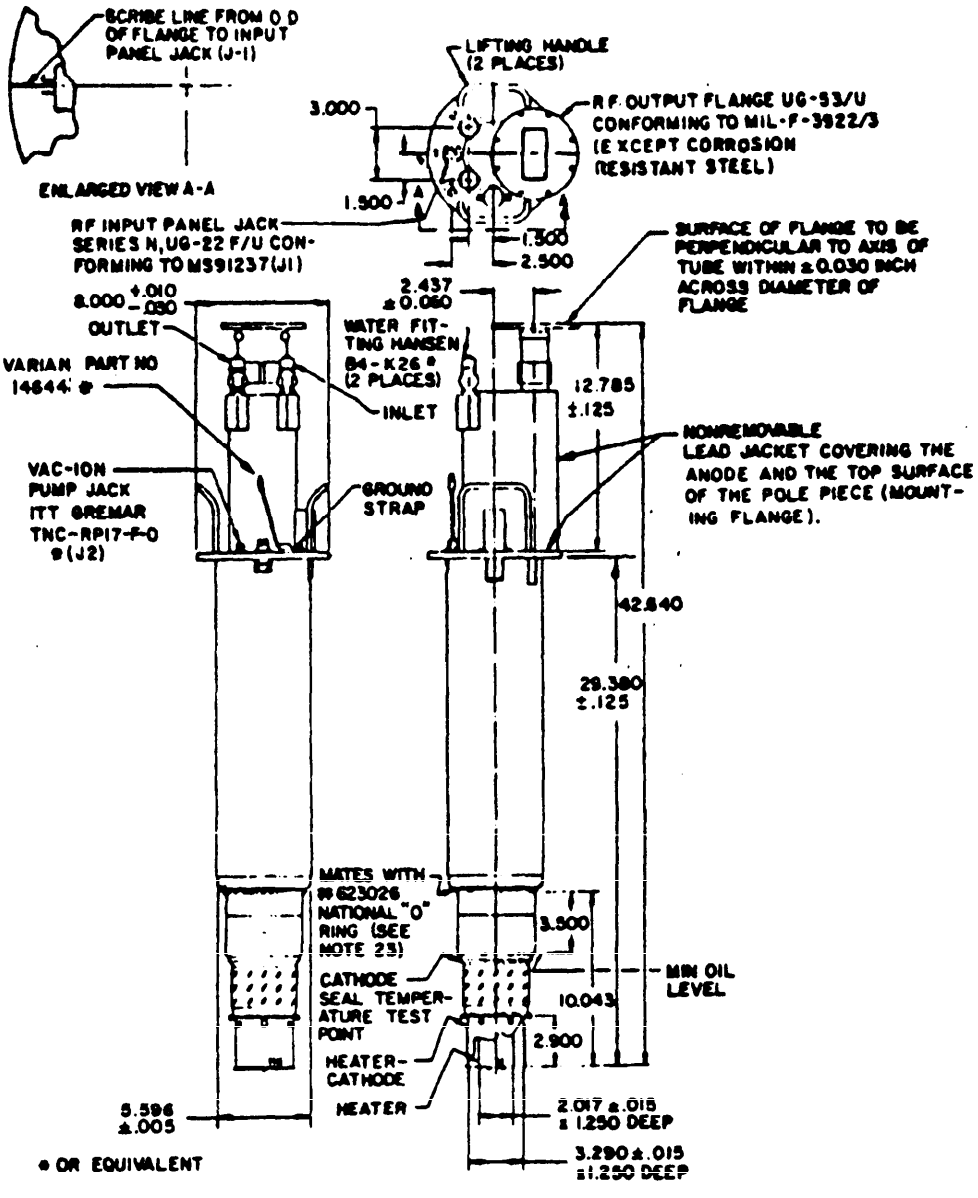
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NOTES:
1. All dimensions are in inches.
2. Those without limits are nominal and shall not be used for acceptance inspection.

FIGURE 1. Outline drawing of electron tube type 8196.

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